ROBERT GORDON UNIVERSITY ABERDEEN

INFORMATICS INSTITUTE OF TECHNOLOGY

BSc(Hons) Artificial Intelligence and Data Science

Module: 2601 OOD

Use case diagram, Use case descriptions and Activity diagrams.

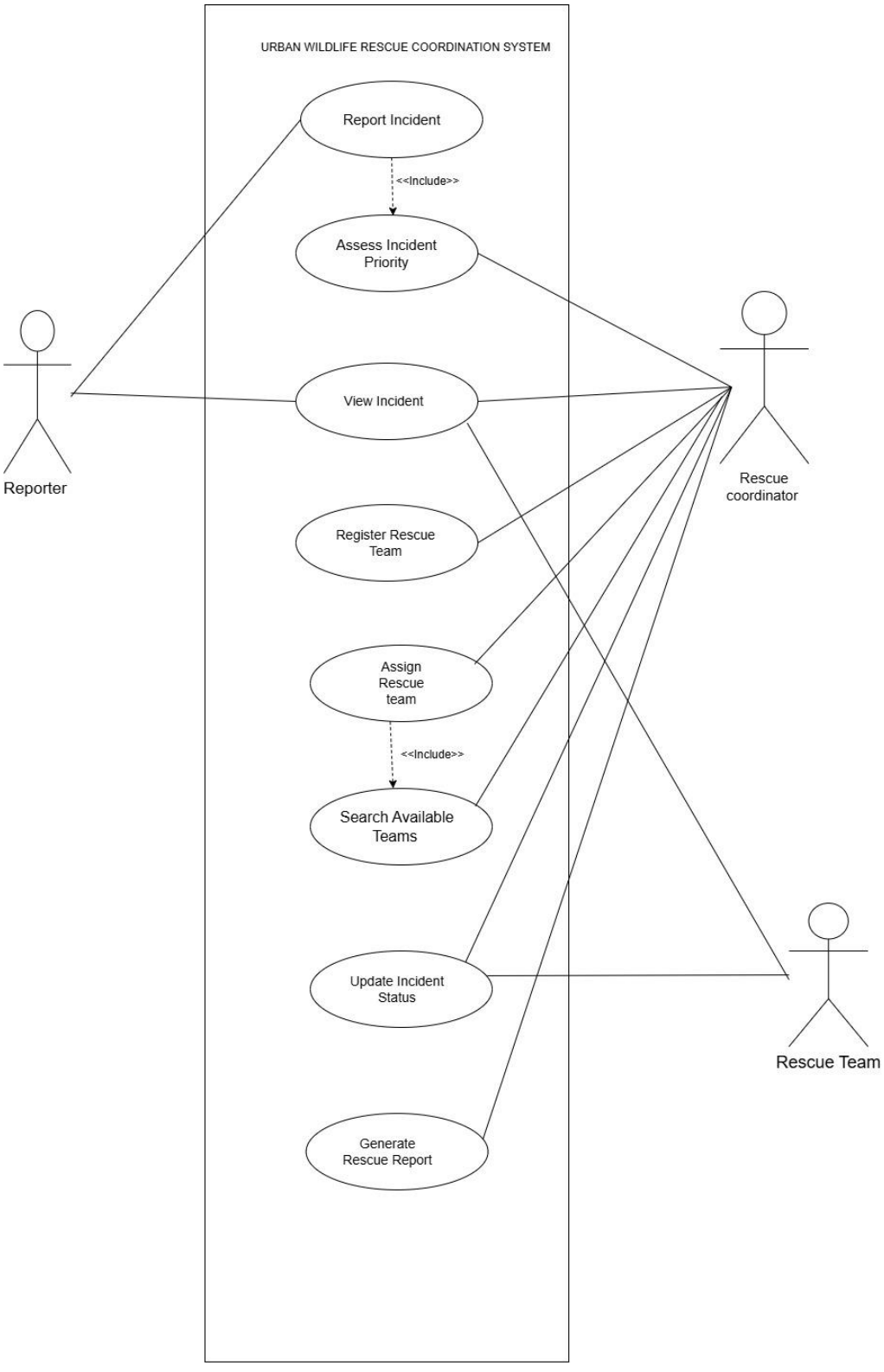
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USE CASE DIAGRAM



Main use cases

	Use case	Actor(s)	Relationship
1	Report incident	Reporter	Includes Assess incident priority
2	Assess incident priority	Rescue coordinator	Included by Report incident
3	Search available teams	Rescue coordinator	Included by Assign rescue team
4	Assign rescue team	Rescue coordinator	Includes Search available teams
5	Register rescue team	Rescue coordinator	None
6	View incident	Reporter, Rescue coordinator,Rescue Team	None
7	Update incident status	Rescue coordinator, Rescue team	None
8	Generate rescue report	Rescue coordinator	None

Use case descriptions

1. Report incident

Section	Main
Use case	Report incident
Actor(s)	Reporter
Purpose	To let a member of the public or staff report an animal that needs rescue.
Overview	This use case begins when a Reporter wants to report an injured, trapped, displaced or abandoned animal. The system checks the details, gives the incident a unique ID, works out its priority and records it as pending. This use case ends when the Reporter has been given the incident ID.
Preconditions	The Reporter has information about an animal that needs rescue.
PostConditions	The incident is recorded with a unique ID, a priority and the status Pending Assignment.

Typical Course of Events

Actor Action	System Response
1. Reporter provides the incident details (animal type, condition, location, number of animals, immediate danger, special requirements, contact details).	2. System checks the details.
	3. System assigns a unique incident ID.
	4. System performs the Assess Incident Priority use case.
	5. System records the incident with the status Pending Assignment.
	6. System gives the incident ID to the Reporter.

Alternative Courses

- Line 2: The details are invalid or missing (for example an invalid animal type or no location). The system tells the Reporter what is wrong and the use case returns to line 1.

2. Assess incident priority

Section	Main
Use case	Assess incident priority
Actor(s)	Rescue coordinator
Purpose	To rank an incident so the most urgent ones are handled first.
Overview	This use case begins when an incident has been reported or the Rescue coordinator asks for it to be reassessed. The system scores the animal's condition, the immediate danger, and the number of animals and location risk, then combines them into a priority category. This use case ends when the incident has a priority of Critical, High, Medium or Low.
Preconditions	The incident has been reported.
Post conditions	The incident has a priority category and its place in the priority order. If the information was invalid, no priority is set.

Typical Course of Events

Actor Action	System Response

1. Rescue coordinator asks for the priority of an incident to be assessed.	2. System checks the information needed to assess the priority.
	3. System scores the animal condition, the immediate danger, and the number of animals and location risk at the same time.
	4. System calculates the total priority score and sets the priority category (Critical, High, Medium or Low).
	5. System places a Critical incident ahead of all other incidents. Any other incident is ranked by its score.
	6. System presents the priority to the Rescue coordinator.

Alternative Courses

- Line 2: The information needed is invalid or missing. The system tells the Rescue coordinator and the use case ends without a priority being set.

3. Search available teams

Section	Main
Use case	Search available teams
Actor(s)	Rescue coordinator
Purpose	To find the rescue teams that are suitable for a pending incident.
Overview	This use case begins when the Rescue coordinator wants to find a team for a pending incident. The system compares what the incident needs with each team's availability, animal categories, skills, equipment and workload. This use case ends when the suitable teams have been presented, or when the Rescue coordinator has been told that none is suitable.
Preconditions	At least one incident is Pending Assignment, and rescue teams have been registered.
Post conditions	The suitable teams have been presented to the Rescue coordinator. No incident or team status is changed.

Typical Course of Events

Actor Action	System Response
1. Rescue coordinator chooses a pending incident.	2. System identifies the animal type, rescue skills and equipment the incident needs.

	3. System selects the teams that are currently Available.
	4. System keeps the teams that can handle the animal type and have the required skills and equipment.
	5. System removes the teams that are already at their maximum workload.
	6. System ranks the remaining teams by suitability.
	7. System presents the suitable teams to the Rescue coordinator.

Alternative Courses

- Line 6: No suitable team remains. The system tells the Rescue coordinator, the incident stays in Pending Assignment, and the use case ends.

4. Assign rescue team

Section	Main
Use case	Assign rescue team
Actor(s)	Rescue coordinator
Purpose	To assign the most suitable available rescue team to a pending incident.
Overview	This use case begins when the Rescue coordinator decides to assign a team to a pending incident. The coordinator chooses a team from the suitable teams, and the system confirms the team can take the job and records the assignment. This use case ends when the incident and the team are both Assigned.
Preconditions	At least one incident is Pending Assignment, and rescue teams have been registered.
Post conditions	The incident and the team have the status Assigned, and the incident is added to the team's workload. If no team could be assigned, the incident stays Pending Assignment.

Typical Course of Events

Actor Action	System Response
1. Rescue coordinator chooses a pending incident to assign.	2. System performs the Search Available Teams use case.
3. Rescue coordinator chooses a team from the suitable teams.	4. System checks that the team is still Available.
	5. System checks that the team has the equipment the incident requires.
	6. System assigns the team to the incident.
	7. System sets the incident and the team to Assigned and adds the incident to the team's workload.
	8. System confirms the assignment to the Rescue coordinator.

Alternative Courses

- Line 2: No suitable team is found. The incident stays in Pending Assignment and the use case ends.
- Line 4: The team is no longer available. The system tells the Rescue coordinator and the use case returns to line 3.
- Line 5: The team does not have the required equipment. The system tells the Rescue coordinator and the use case returns to line 3.

5. Register rescue team

Section	Main
Use case	Register rescue team
Actor(s)	Rescue coordinator
Purpose	To add a new rescue team so it can be matched to incidents.
Overview	This use case begins when the Rescue coordinator wants to add a new rescue team. The coordinator provides the team's details, and the system checks them and adds the team. This use case ends when the team has been registered.
Preconditions	The Rescue coordinator has the details of the new team.
Post conditions	The new team is registered and can be matched to incidents.

Typical Course of Events

Actor Action	System Response
1. Rescue coordinator provides the team details (team ID, members, animal categories, rescue skills, equipment, current status, maximum simultaneous incidents).	2. System checks the details.
	3. System checks that the team ID is not already in use.
	4. System adds the team to the organisation's rescue teams.
	5. System confirms the registration to the Rescue coordinator.

Alternative Courses

- Line 2: The team details are invalid or missing (for example an invalid animal category). The system tells the Rescue coordinator and the use case returns to line 1.
- Line 3: The team ID is already in use. The system tells the Rescue coordinator and the use case returns to line 1.

6. View incident

Section	Main
Use case	View incident
Actor(s)	Reporter, Rescue coordinator, Rescue team
Purpose	To let an actor see the details and current status of a reported incident.
Overview	This use case begins when a Reporter, Rescue coordinator or Rescue team wants to see an incident. A Reporter checks the progress of a report, the Rescue coordinator reviews any incident, and a Rescue team looks at its incident before updating its status. The actor identifies the incident, and the system finds it and presents its details. This use case ends when the incident details have been presented.
Preconditions	The incident has been reported, and the actor knows its incident ID or can pick it from the list of incidents.
Post conditions	The incident details have been presented. Nothing is changed.

Typical Course of Events

Actor Action	System Response
1. Reporter, Rescue coordinator or Rescue team selects the incident to view.	2. System finds the incident.
	3. System presents the incident details (animal type, condition, location, priority, status and assigned team).

Alternative Courses

- Line 2: No incident matches the one selected. The system tells the actor and the use case returns to line 1.

7. Update incident status

Section	Main
Use case	Update incident status
Actor(s)	Rescue coordinator, Rescue team
Purpose	To record the progress of a rescue so the incident status is always up to date.
Overview	This use case begins when a Rescue team or the Rescue coordinator wants to record progress on an incident. The actor chooses the incident and its new status, and the system checks the change is allowed and records it. This use

	case ends when the new status has been recorded and, for a Completed incident, the team is Available again.
Preconditions	The incident has been assigned to a team, its status is Assigned or In Progress, and the actor knows the incident (for example after viewing it).
Post conditions	The incident has its new status. If the status is Completed, the assigned team is Available again and its workload is reduced.

Typical Course of Events

Actor Action	System Response
1. Rescue team or Rescue coordinator selects the incident.	2. System finds the incident.
3. Rescue team or Rescue coordinator chooses the new status (In Progress or Completed).	4. System checks that the status change is allowed.
	5. System updates the incident status.
	6. If the new status is Completed, the system makes the assigned team Available again and reduces its workload.
	7. System confirms the update.

Alternative Courses

- Line 2: No incident matches the one selected. The system tells the actor and the use case returns to line 1.
- Line 4: The change is not allowed (for example Assigned straight to Completed). The system rejects the change and the use case returns to line 3.

8. Generate rescue report

Section	Main
Use case	Generate rescue report
Actor(s)	Rescue coordinator
Purpose	To produce a report of the completed and pending rescue incidents.
Overview	This use case begins when the Rescue coordinator asks for the rescue report. The system gathers all reported incidents, separates the completed ones from the pending ones (those not yet Completed) and totals each group. This use case ends when the report has been presented and saved.
Preconditions	The Rescue coordinator has access to the system.
Post conditions	The report has been presented to the Rescue coordinator and saved.

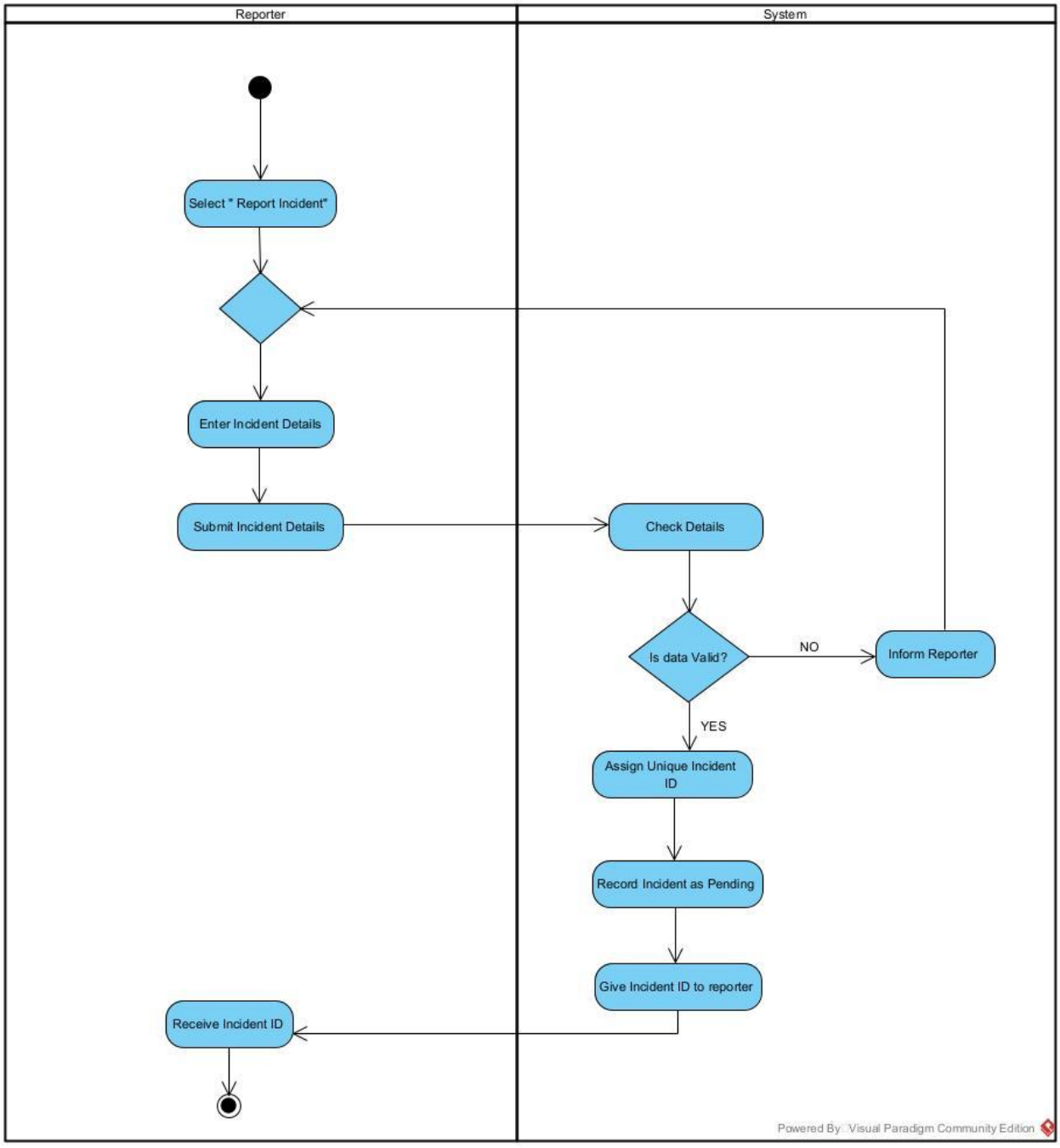
Typical Course of Events

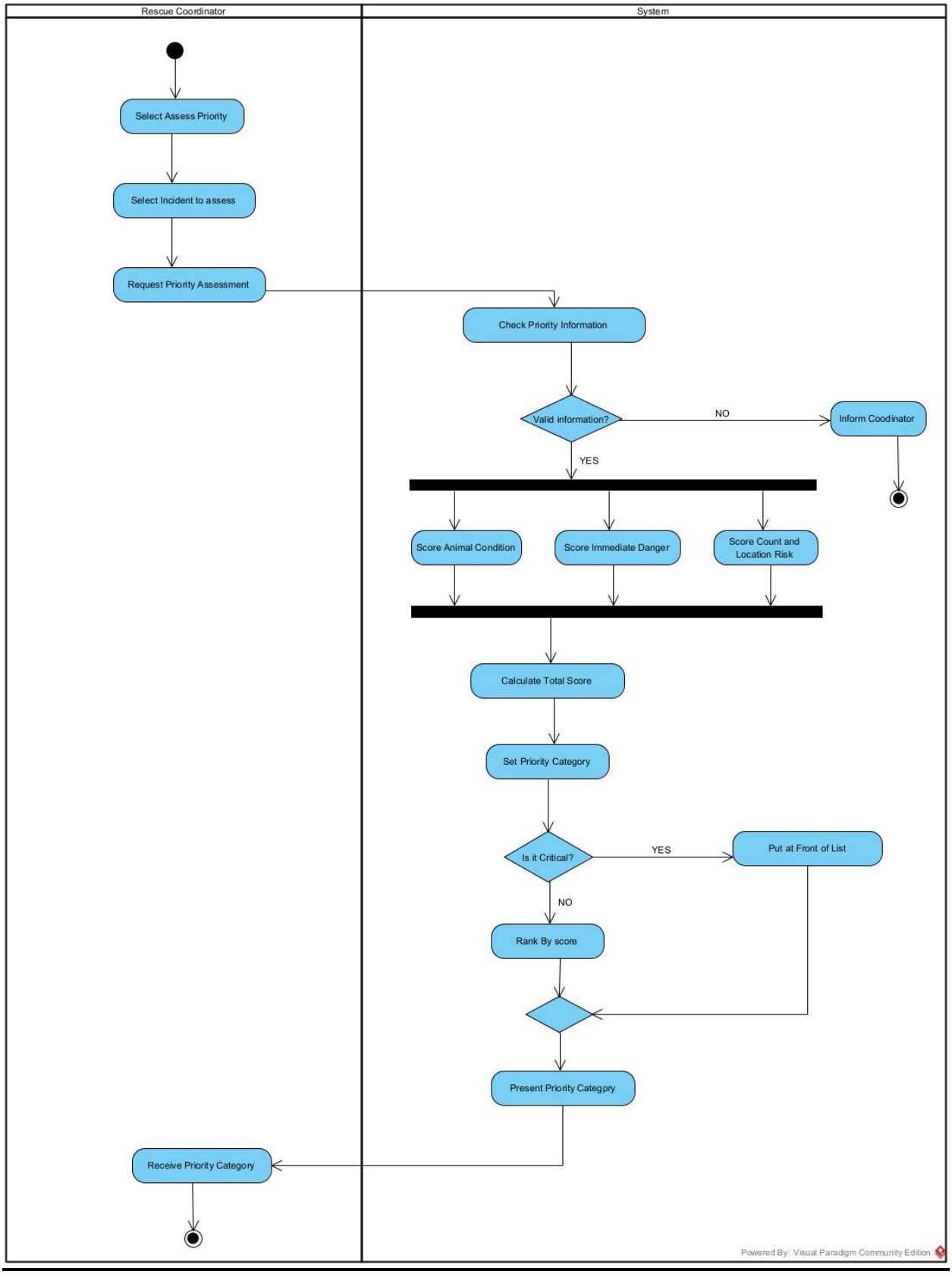
Actor Action	System Response
1. Rescue coordinator asks for the rescue report.	2. System gathers all reported incidents.
	3. System groups the incidents into completed and pending.
	4. System totals the incidents in each group.
	5. System presents the report to the Rescue coordinator.
	6. System saves the report.

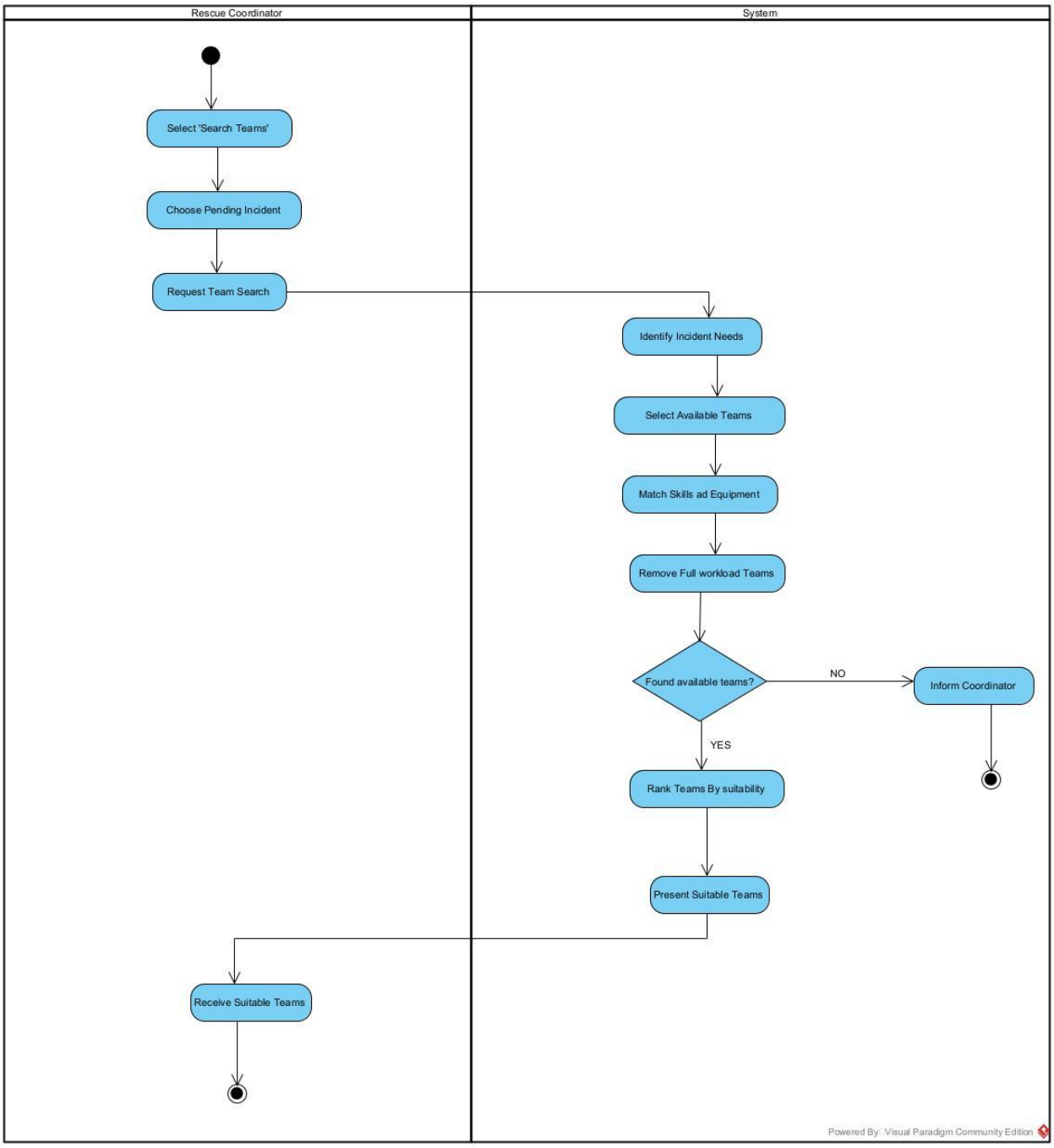
Alternative Courses

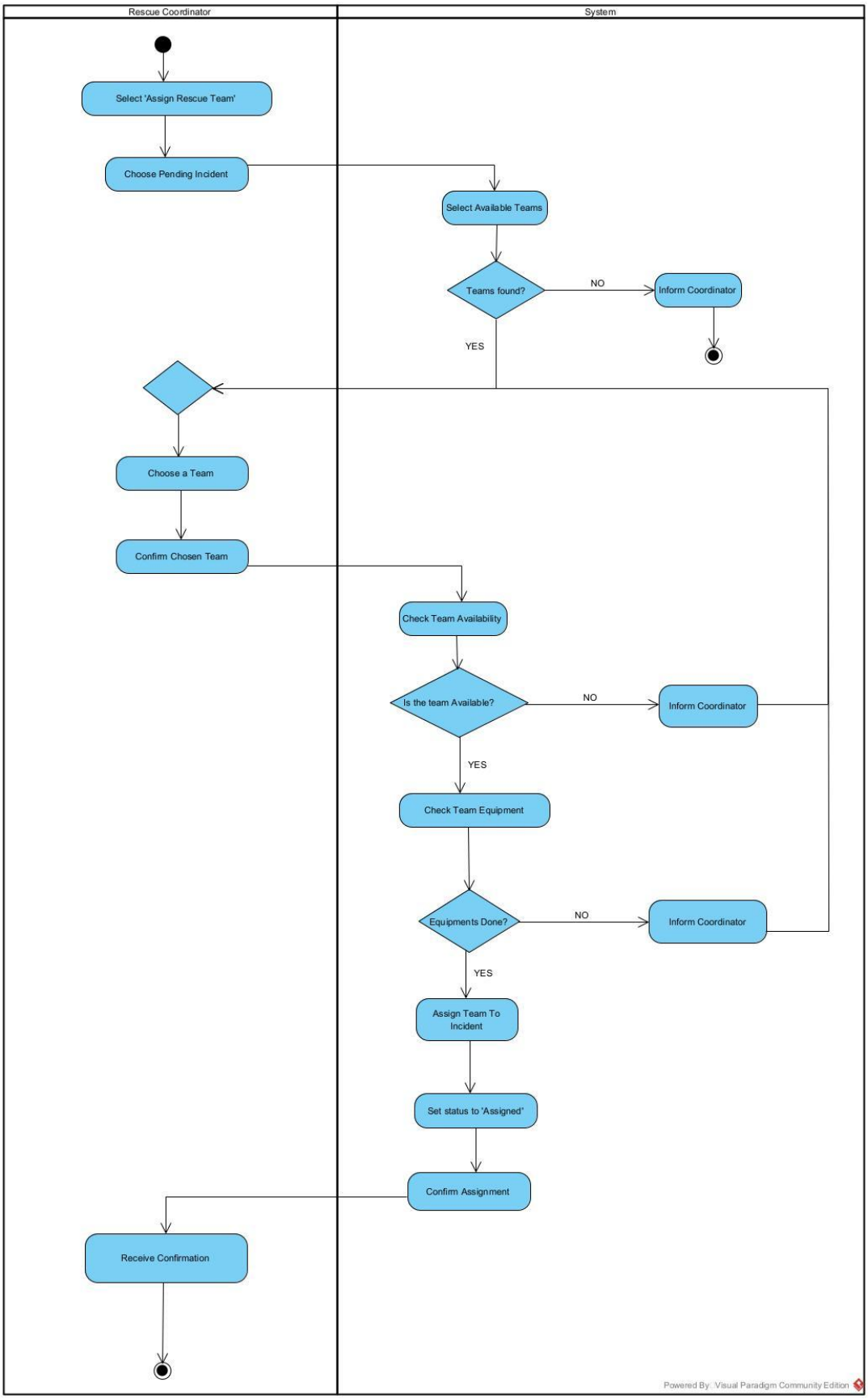
- Line 2: No incidents have been reported. The system tells the Rescue coordinator and the use case ends.

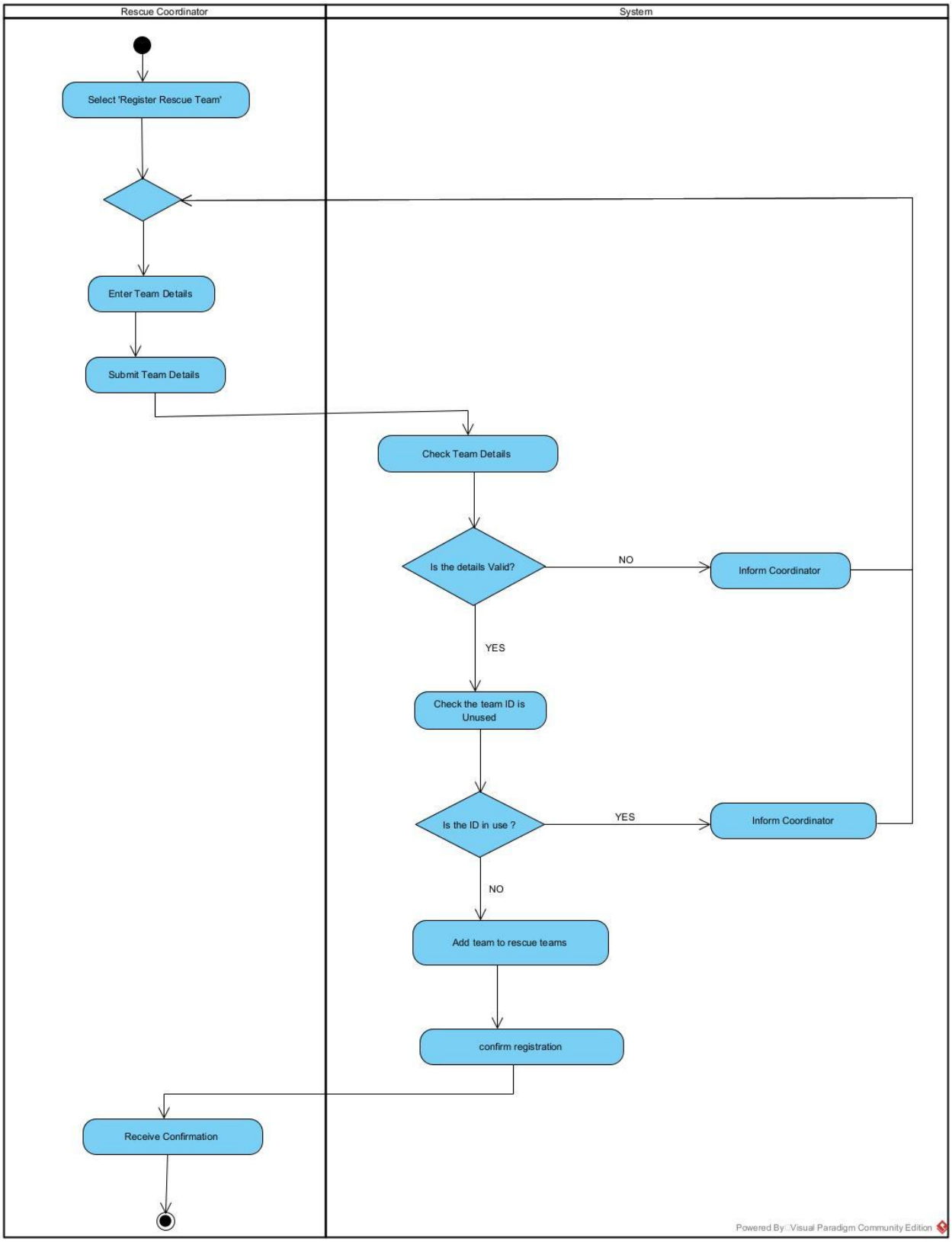
Activity Diagrams

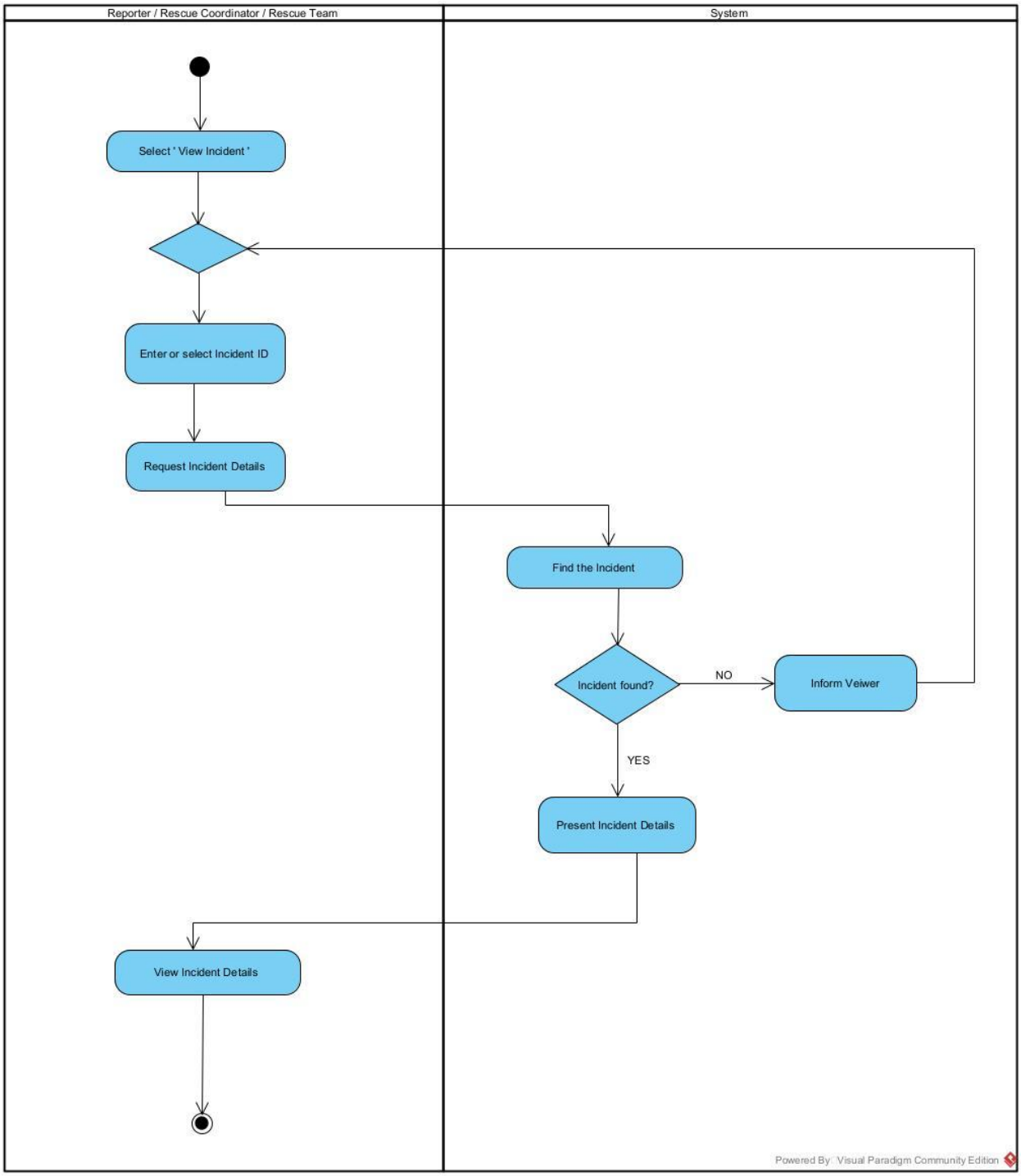


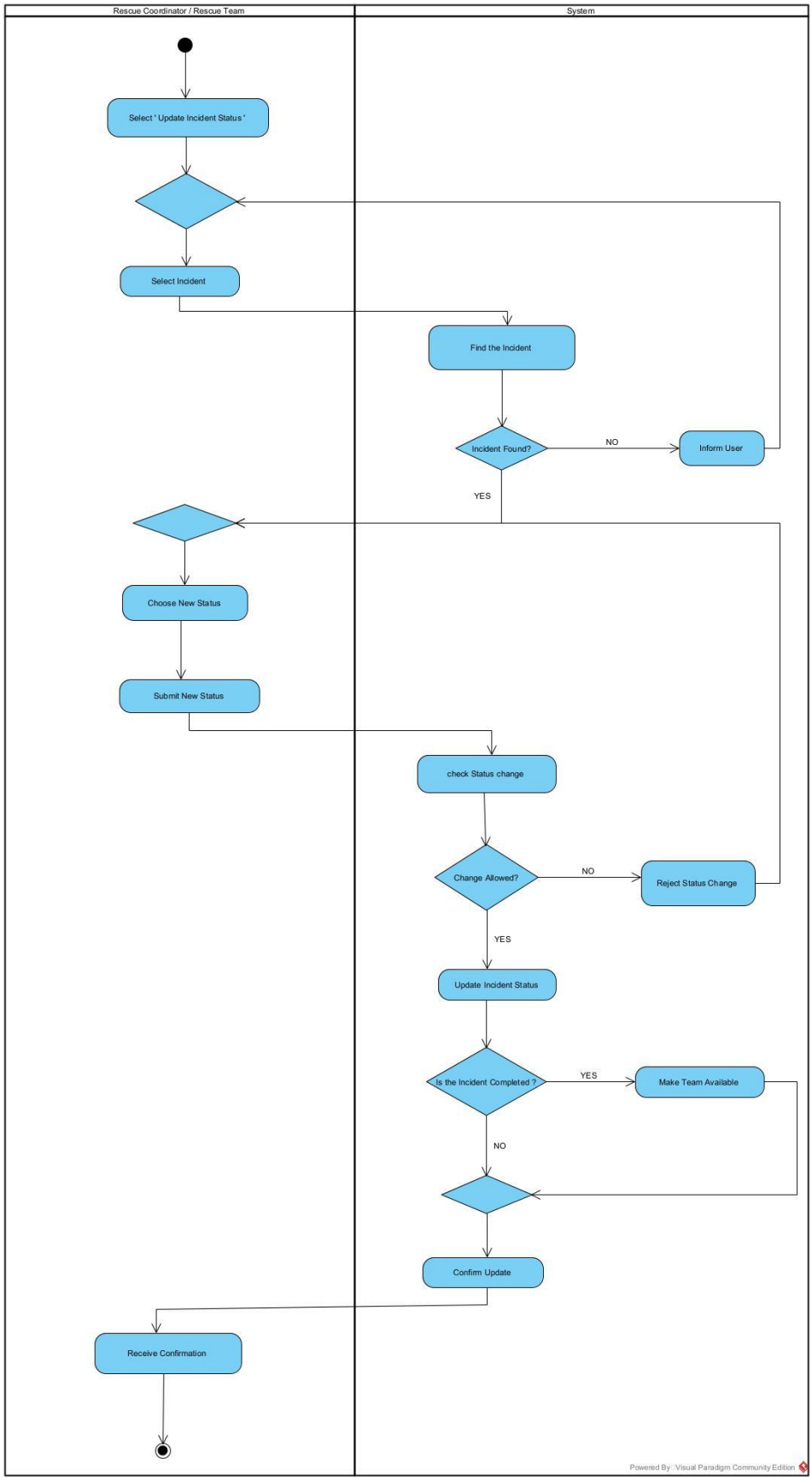

Rescue Coordinator
System
Select Assess Priority
Select Incident to assess
Request Priority Assessment
Check Priority Information
Valid information?
NO
Inform Coodinator
YES
Score Animal Condition
Score Immediate Danger
Score Count and Location Risk
Calculate Total Score
Set Priority Category
Is it Critical?
YES
Put at Front of List
NO
Rank By score
Present Priority Categpry
Receive Priority Category
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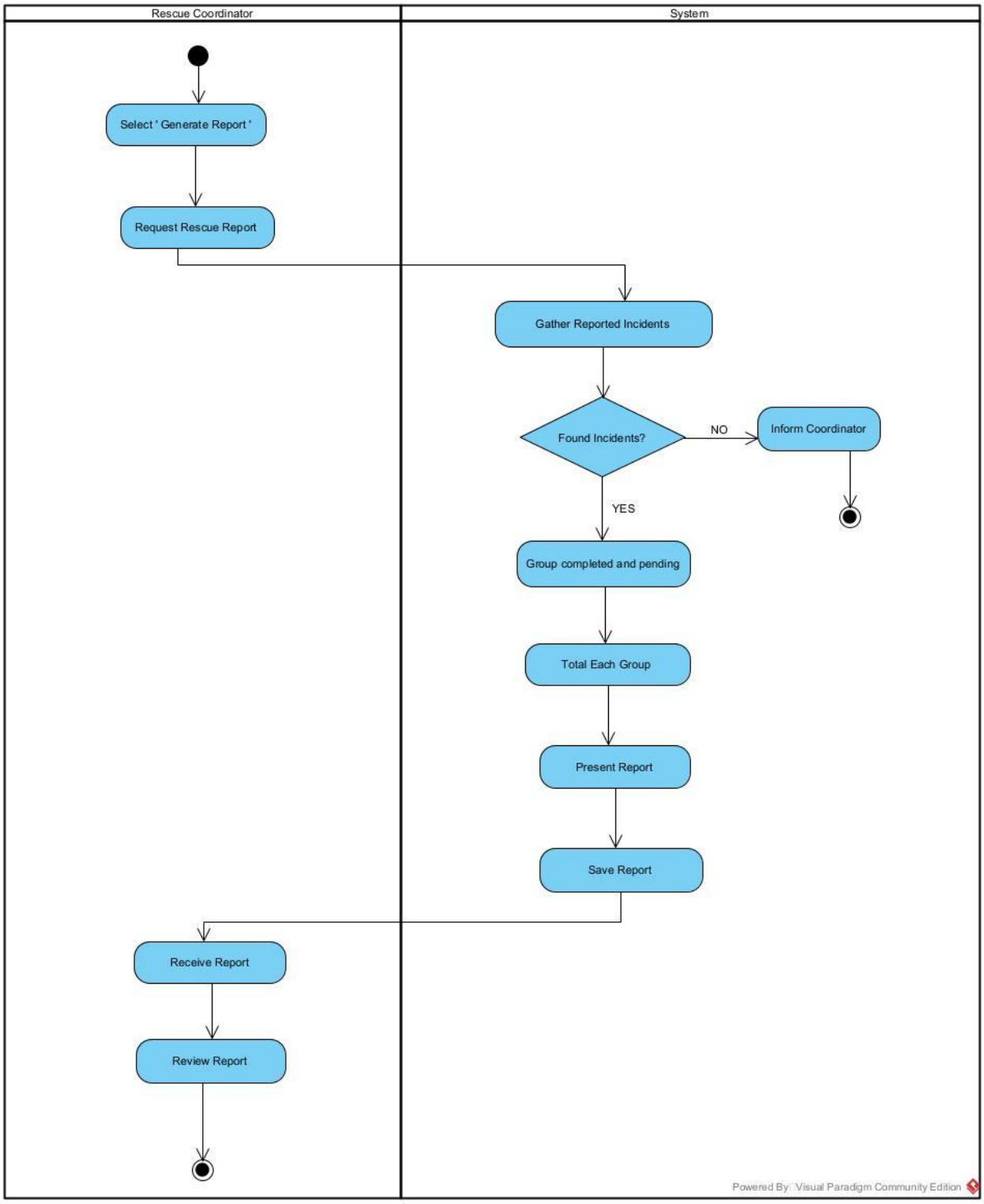

Rescue Coordinator
System
Select 'Search Teams'
Choose Pending Incident
Request Team Search
Identify Incident Needs
Select Available Teams
Match Skills ad Equipment
Remove Full workload Teams
Found available teams?
NO
Inform Coordinator
YES
Rank Teams By suitability
Present Suitable Teams
Receive Suitable Teams
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Rescue Coordinator
System
Select 'Assign Rescue Team'
Choose Pending Incident
Select Available Teams
Teams found?
NO
Inform Coordinator
YES
Choose a Team
Confirm Chosen Team
Check Team Availability
Is the team Available?
NO
Inform Coordinator
YES
Check Team Equipment
Equipments Done?
NO
Inform Coordinator
YES
Assign Team To Incident
Set status to 'Assigned'
Confirm Assignment
Receive Confirmation
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Rescue Coordinator
System
Select 'Register Rescue Team'
Enter Team Details
Submit Team Details
Check Team Details
Is the details Valid?
NO
Inform Coordinator
YES
Check the team ID is Unused
Is the ID in use ?
YES
Inform Coordinator
NO
Add team to rescue teams
confirm registration
Receive Confirmation
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Reporter / Rescue Coordinator / Rescue Team
System
Select ' View Incident '
Enter or select Incident ID
Request Incident Details
Find the Incident
Incident found?
NO
Inform Veiwer
YES
Present Incident Details
View Incident Details
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Rescue Coordinator / Rescue Team
System
Select ' Update Incident Status '
Select Incident
Find the Incident
Incident Found?
NO
Inform User
YES
Choose New Status
Submit New Status
check Status change
Change Allowed?
NO
Reject Status Change
YES
Update Incident Status
Is the Incident Completed ?
YES
Make Team Available
NO
Confirm Update
Receive Confirmation
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Rescue Coordinator
System
Select ' Generate Report '
Request Rescue Report
Gather Reported Incidents
Found Incidents?
NO
Inform Coordinator
YES
Group completed and pending
Total Each Group
Present Report
Save Report
Receive Report
Review Report
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